The Master Mechanist's Compendium of Clinical Physiology

Advanced Diagnostics & Hemodynamic Calibration for the Acute Care Practitioner

Operational Manual v.1.0 // Authorized Personnel Only

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SYSTEM DIAGNOSTIC SEQUENCE

1. Module I: The Cardiac Engine – Pump Integrity & Pressure Analysis
2. Module II: Chemical Calibration – Acid-Base Equilibrium Algorithms
3. Module III: The Ventilation System – Bellows Dynamics & Pressure Alarms
4. Module IV: Operational Priority – Triage Protocols & Hierarchy of Needs

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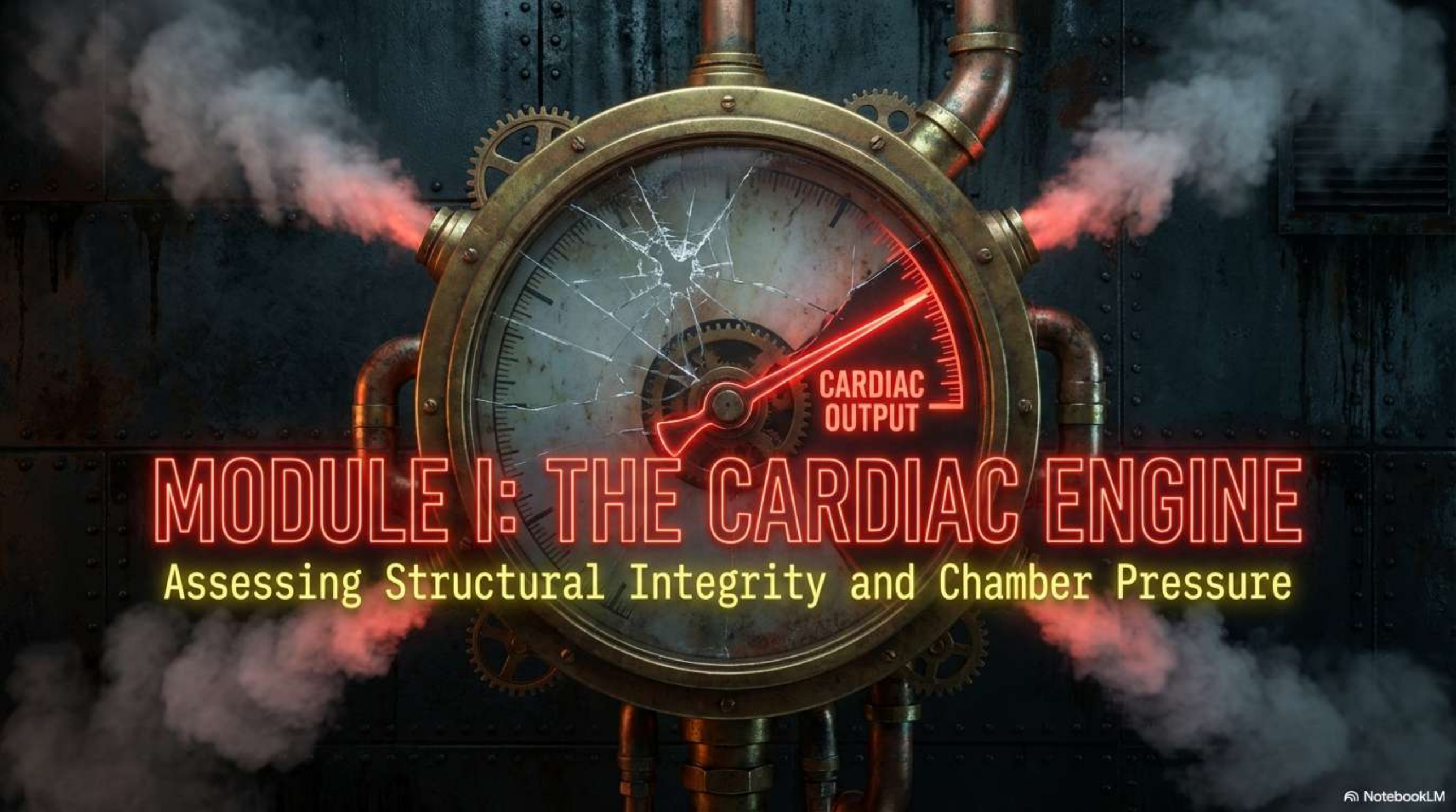

MODULE I: THE CARDIAC ENGINE

Assessing Structural Integrity and Chamber Pressure

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Structural Integrity Breach: Troponin

Core Concept: The Gold Standard for Myocardial Infarction (MI).

Data Threshold: > 0.5 ng/mL

Significance: CELL DEATH CONFIRMED.

Note: This is not a warning light. It is a confirmation that the machinery is physically breaking down. Metal shards are in the oil.

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Chamber Pressure Analysis: BNP

Mechanism: B-Type Natriuretic Peptide is secreted in response to excessive STRETCHING of the ventricular walls.

Thresholds:
- < 100: Normal Operation
- > 100: Critical Indicator of Congestive Heart Failure (CHF) ⚠

The Analogy: The “Big Stretched Ventricle” Gauge.

Clinical Pearl: High BNP confirms dyspnea is Cardiac (Pump Failure), not Pulmonary.

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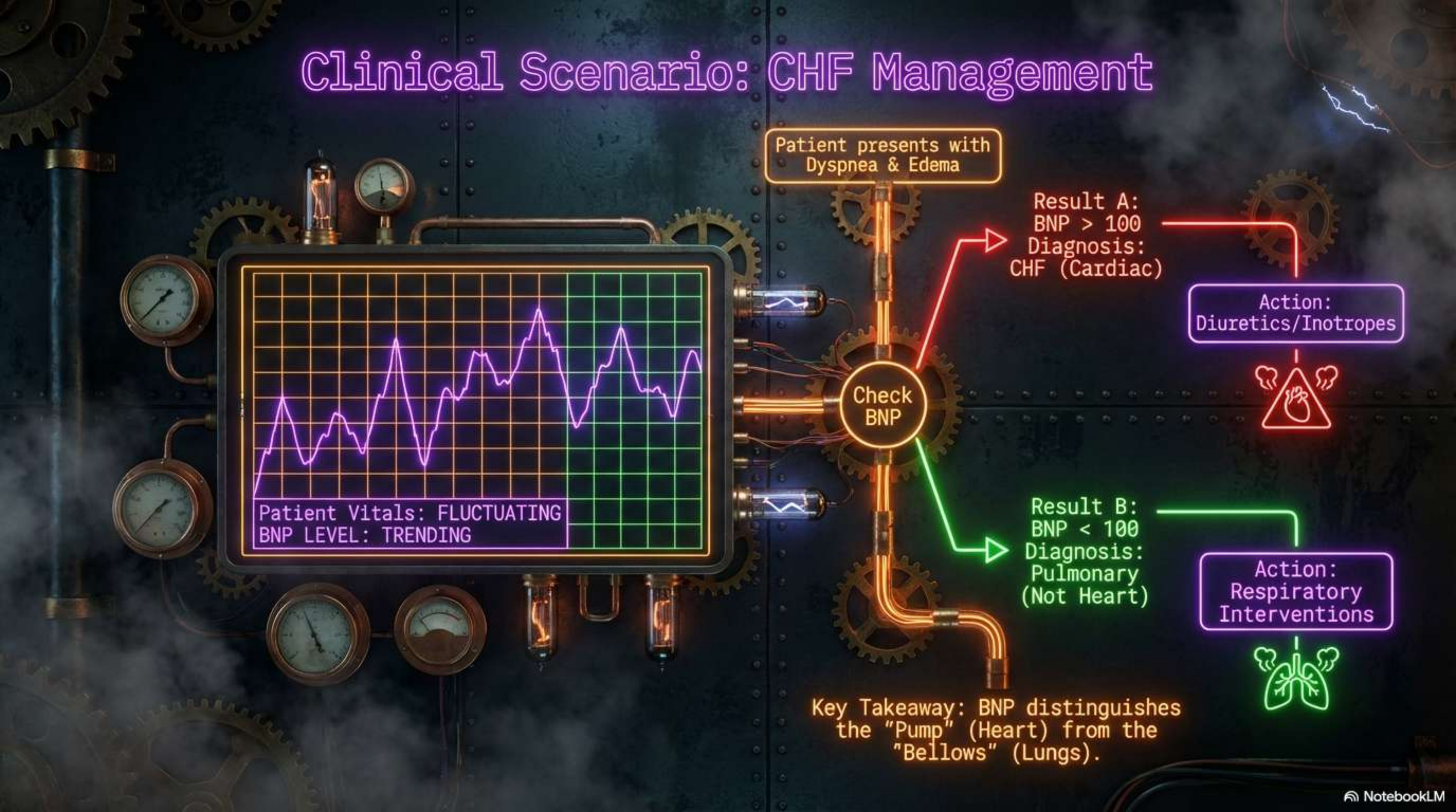
Clinical Scenario: CHF Management



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MODULE II: CHEMICAL CALIBRATION

The Acid-Base Equilibrium & The Rule of B's

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The Golden Rule of Calibration: The Rule of B's

The Rule: If the pH and the Bicarb (HCO3) are BOTH moving in the same direction, it is MetaBOLIC.

Examples:
- pH ↓ + HCO3 ↓ = Metabolic Acidosis
- pH ↑ + HCO3 ↑ = Metabolic Alkalosis
- pH ↓ + HCO3 ↑ = Respiratory Acidosis (Opposite Directions)

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Systemic Synchronization

Axiom: As the pH goes, so goes the patient.

EXCEPTION: POTASSIUM (K+)

High pH = Low Potassium (Hypokalemia)
Low pH = High Potassium (Hyperkalemia) [DANGER]

High pH (Alkalosis) = System Overload

Low pH (Acidosis) = System Shutdown

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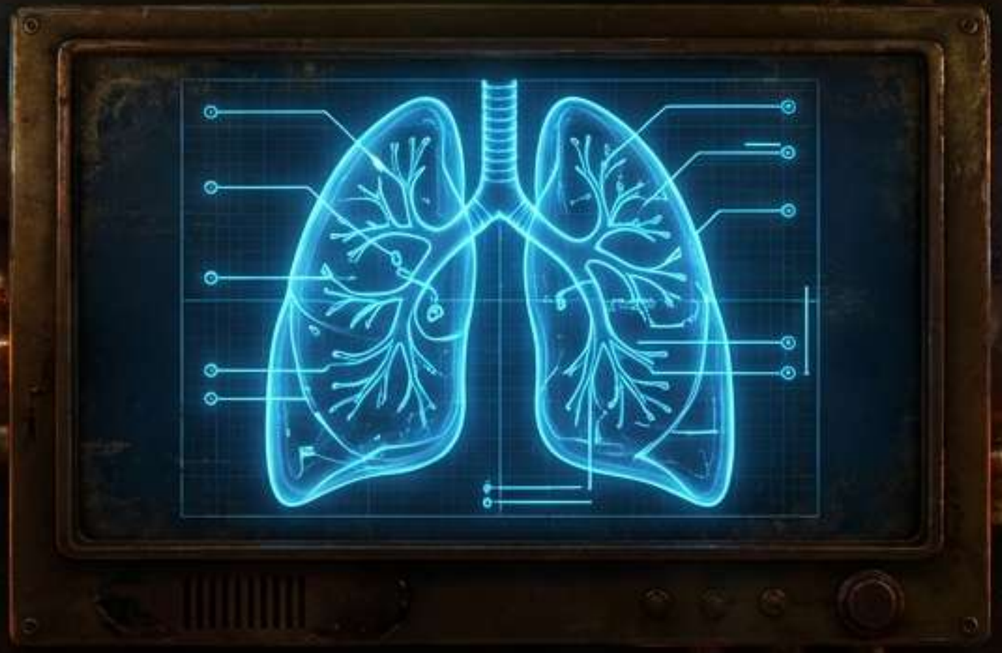
Diagnostic Algorithm: Step 1

Is the pathology in the LUNGS?



Is it the Lungs? -> YES -> Respiratory Imbalance.

Is it the Lungs? -> NO -> Metabolic Imbalance.

Note: If the scenario is complex or unknown, default setting is Metabolic Acidosis.

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Diagnostic Algorithm: Step 2 - Ventilation Dynamics

Concept: It is not about Rate; it is about SaO_2 and CO_2 Clearance.

Logic:

- Over-ventilating (Blowing off CO_2) -> Respiratory Alkalosis (Alkalosis = Over).
- Under-ventilating (Retaining CO_2) -> Respiratory Acidosis (Acidosis = Under).

Interventions:

- Ambu-bag the Acidosis.
- Paper-bag the Alkalosis.

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Metabolic Acidosis: The Default Failure Mode

Primary Causes:

1. Renal Failure (Filter Blockage).
2. Severe Diarrhea (Loss of base from lower GI).

Hallmark Sign: MacKussmaul Respirations.

- Deep, labored breathing pattern.
- “Mac” = Metabolic ACidosis.

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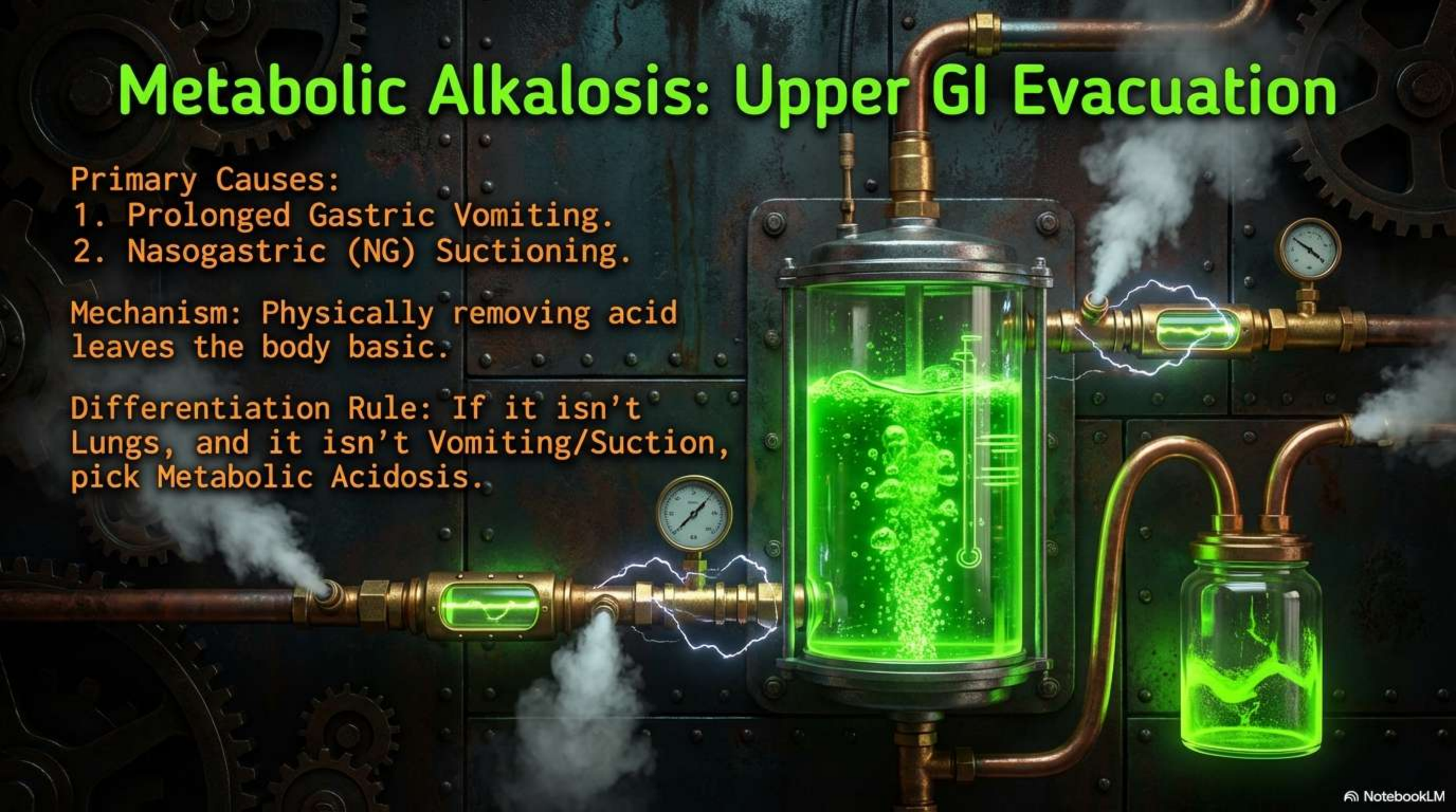
Metabolic Alkalosis: Upper GI Evacuation

Primary Causes:
1. Prolonged Gastric Vomiting.
2. Nasogastric (NG) Suctioning.

Mechanism: Physically removing acid leaves the body basic.

Differentiation Rule: If it isn't Lungs, and it isn't Vomiting/Suction, pick Metabolic Acidosis.



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Deep Dive: The Alcohol-Alkalosis Paradox

Pathophysiology Chain:

1. Intoxication/Withdrawal -> Excessive Excessive Vomiting -> Volume Depletion -> Metabolic Alkalosis.

2. Thiamine Deficiency -> Poor Nutrition + High Metabolic Demand -> Wernicke-Korsakoff Syndrome.

CRITICAL INTERVENTION:
Administer Thiamine (Vitamin B1) BEFORE or WITH Glucose to prevent permanent brain damage (Wernicke's Encephalopathy).

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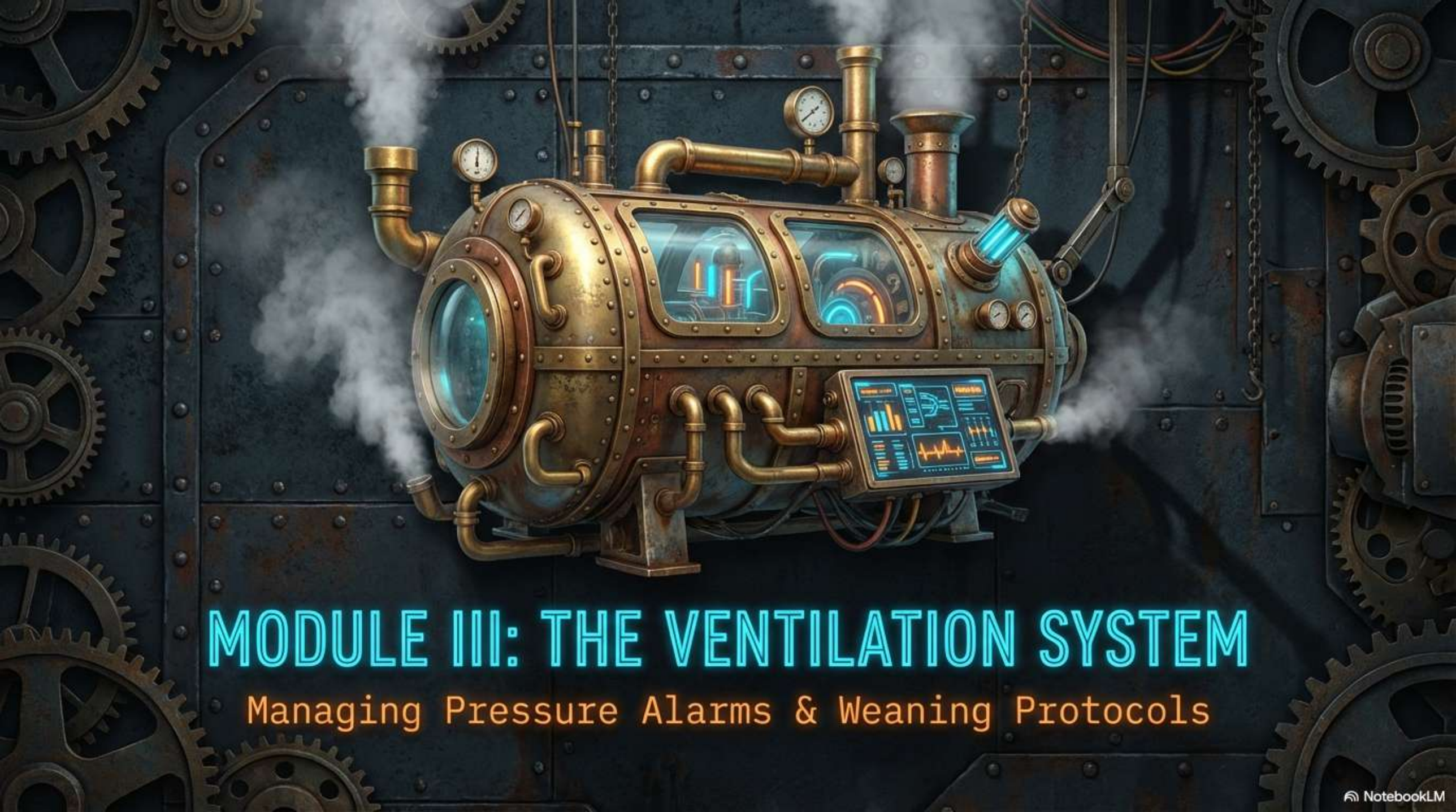

MODULE III: THE VENTILATION SYSTEM

Managing Pressure Alarms & Weaning Protocols

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High Pressure Alarm = INCREASED Resistance

Logic: The machine is pushing hard but cannot get air in. The path is blocked.

Obstruction Causes:

1. Kinks in the tubing.
2. Water condensation in the tube.
3. Mucus secretions in the airway.

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The Obstruction Resolution Sequence

1. UNKINK: Check the tubing for bends.
2. EMPTY: Drain water from dependent loops.
3. TURN/COUGH/DEEP BREATHE: Mobilize secretions.
4. SUCTION: Only if step 3 fails. (Last Resort).

Warning: Do not deviate from this order.

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Low Pressure Alarm = DECREASED Resistance

Logic: The machine is pushing air and finding zero resistance. The closed loop is broken.

Causes:
1. Main Tubing Disconnection.
2. Oxygen Sensor Disconnection.

Intervention: Reconnect immediately.

Contingency: If tube hits the floor -> Bag the patient -> Call Respiratory Therapy.

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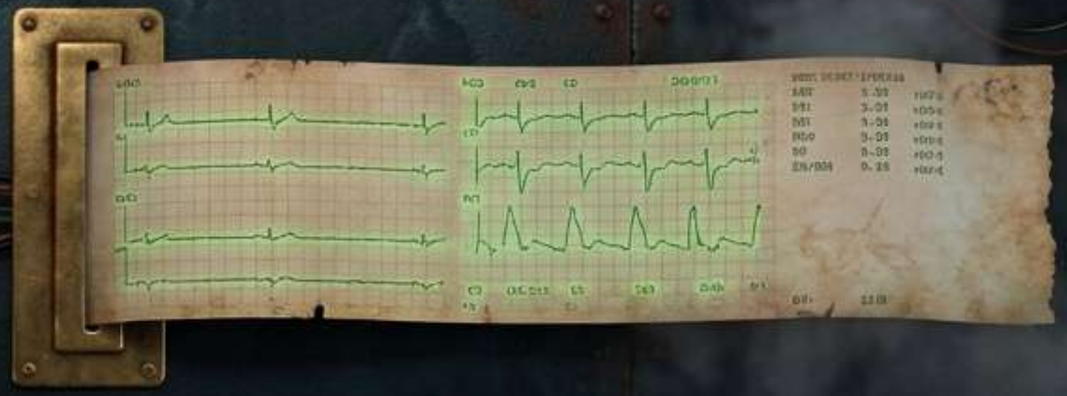

Case Scenario: The Weaning Decision

Scenario: 06:00 AM. Order to wean. Patient ABG shows Respiratory Acidosis.

Analysis:
- Respiratory Acidosis = Under-ventilating.
- The patient is failing to clear CO_2 on current settings.

Action: HOLD THE ORDER. Call the Physician.

Rationale: Patient is not ready. (Only consider weaning if Alkalotic/Over-ventilating).

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MODULE IV: OPERATIONAL PRIORITY

Triage Protocols & The Hierarchy of Needs

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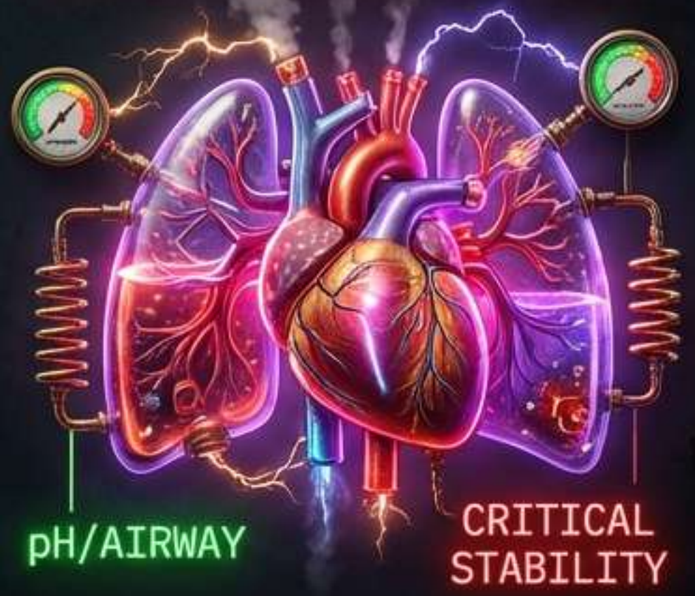
Maslow vs. Physiology

MASLOW

ACUTE MODIFICATION



Standard Rule: Physiological needs come first.

The Modification: In acute crisis, Physiological Stability (pH/Airway) trumps all.

Example: Respiratory Acidosis (Physiological Failure) > Pain (Comfort) > Anxiety (Psych)

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